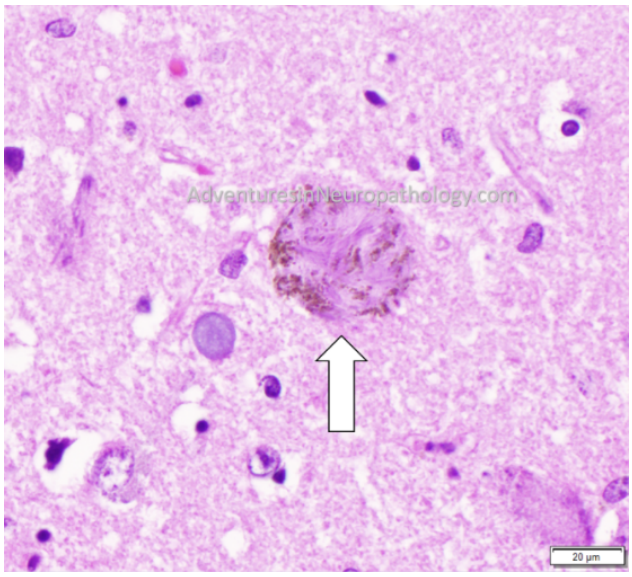
2025 Mitosisphere August Challenge

Part A (MCQ)

Anatomy & Physiology (2 Questions)

Question 1.
Shown below is an image taken from the brain of an individual who suffers from a certain neurodegenerative disease. The region marked by the arrow corresponds to buildups of a mutated version of a certain microtubule associated protein (MAP) that functions in both the structural maintenance of, and transport of nutrients inside neurons.



Which of the following diseases does this individual suffer from?

A. Parkinson's

B. Alzheimer's

C. Schizophrenia

D. Huntington's

E. Creutzfelt-Jakob Syndrome

Question 2.
Anish was a staunch advocate of the excretory system being the best in scioly. Understandably, the entirety of BODS believed that he was wrong and Mr.

Lahmnaj condemned Anish to suffer a complete failure of his innate immune system. As a result, he developed an immense tumor (not cancer) in the cells of his small intestine due to bacteria invading and preventing the action of both Iron regulatory proteins(IRPs) and Iron regulatory elements(IREs) from acting and instead taking up all of the compound that these (IRPs+IREs) act on. Which of the following antibiotics OR ENZYMES could potentially help Anish survive until he repents? His blood type is AB.

a. Lactoferrin

b. Lysozyme

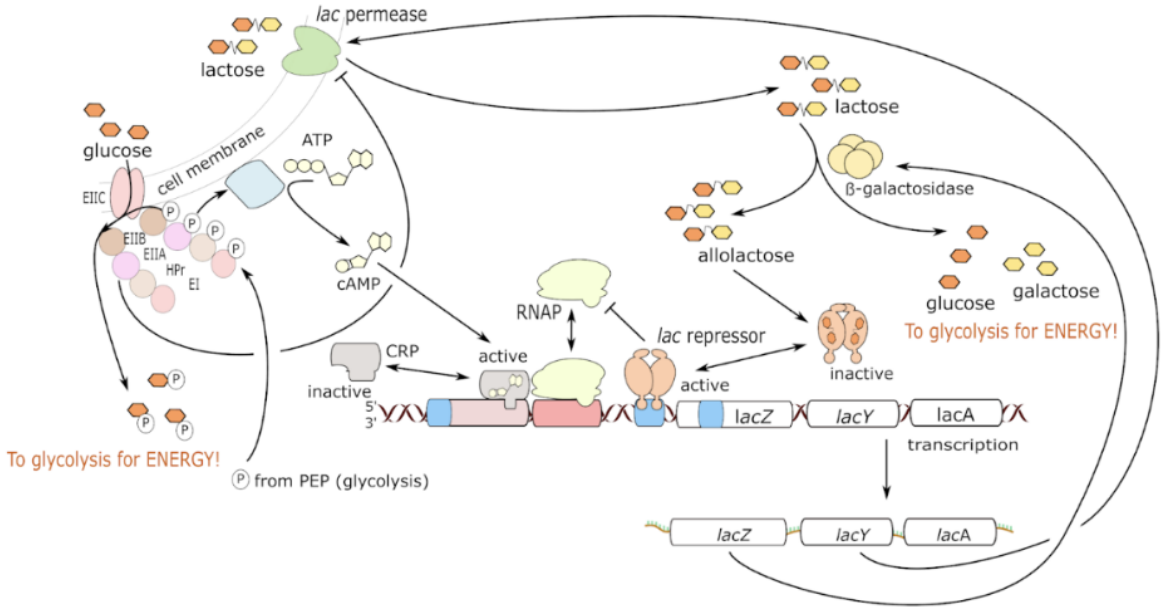
c. Cl–inhibiting Penicillin Concentrate

d. N-terminal Tetracycline

e. AB+-specific aminoglycosides

Cell Biology (2 Questions)

Question 3.
Shown below is a model of the lac operon. The lac operon is under dual positive and negative control, regulated by both internal glucose and lactose concentrations. When glucose levels are high, cAMP levels are low and the cell preferentially catabolizes glucose. When glucose levels are low, cAMP levels are high and the cell resorts to lactose catabolism.



If you were to introduce a competitive inhibitor of cAMP phosphodiesterase into a cell under low glucose and high lactose conditions, how would the levels of

beta-galactosidase expression compare to that of a cell under the same conditions without this inhibitor, and to that of a cell with high glucose and low lactose without this inhibitor, respectively?

A. Higher, Higher

B. Lower, Lower

C. Higher, Lower

D. Lower, Higher

E. Same, Lower

Question 4.
If the trp attenuator sequence was modified to change all the tryptophan codons to tyrosine codons, which of the following changes should you also make to make the operon function again? Select all that apply.

A. Adding a region downstream of the 4th complementary region that is also complementary

B. Adding a region upstream of the 1st complementary region that is also complementary

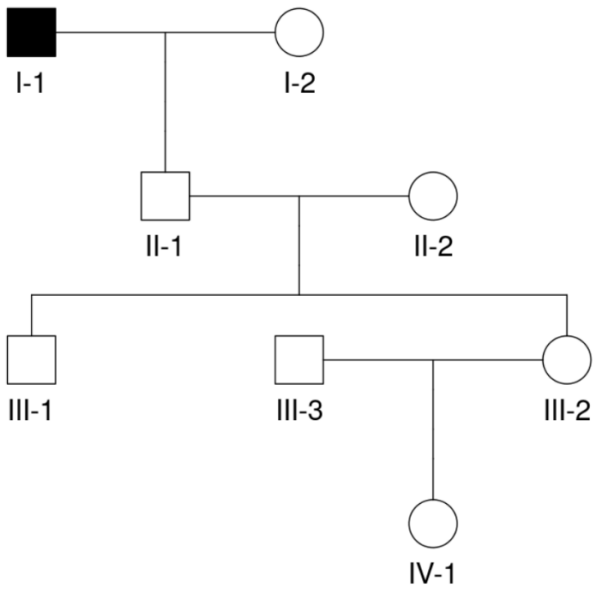
C. Deleting the 4th complementary region

D. Deleting the 1st complementary region

E. Replacing the U-rich attenuator with a G rich attenuator

Genetics (2 Questions)

Question 5.
The pedigree below shows the inheritance of the rare disease Ivanophia, which is characterized by the sudden desire to reincarnate Ivan the Terrible, the First Tsar of Russia from the years 1533 to 1547. There is a 1% chance that a given unaffected person is a carrier. III-2 (Michaelis) and III-3 (Menten) wants to have a child, (IV-1), Enzyme Kinetics, what is the total probability that Enzyme Kinetics is a carrier or affected? (Ivanophia is a autosomal recessive condition)



A. 0.25345

B. 0.34545

C. 0.25375

D. 0.15

E. 0.38125

Question 6.
Ken has been studying the pathway for the synthesis of pigment in a species of garden snail. Ken knows that the pathway for pigment synthesis involves the conversion of a colorless precursor to red pigment by enzyme A, and the conversion of red pigment to yellow pigment by enzyme B. The inheritance pattern for enzyme B is incompletely dominant, where individuals heterozygous for B but having at least 1 A allele are orange. Ken breeds a true breeding red individual with a true breeding white individual. All of the f1 generation is orange. What is the phenotypic ratio you would expect to see if Ken were to self cross the f1 generation?

A. 1 red : 2 orange : 1 yellow

B. 9 yellow : 3 red : 4 white

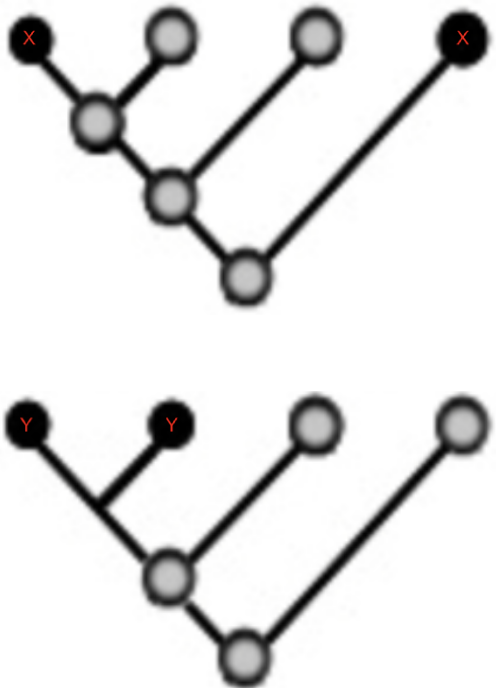
C. 9 yellow : 3 orange : 3 red : 1 white

D. 3 yellow : 6 orange : 3 red :4 white

E. 6 yellow : 3 orange : 3 red : 4 white

Evolution (1 Question)

Question 7.
You are studying the evolution of traits X and Y in marine mammals. You have recorded the data regarding the evolution of the trait below:



Using this data, you conclude that trait X is most likely a _____, while trait Y is most likely a _____.

A. Homoplasy, apomorphy

B. Apomorphy, homoplasy

C. Pleisomorphy, synapomorphy

D. Synapomorphy, pleisomorphy

E. Apomorphy, synapomorphy

Plant Biology (1 Question)

Question 8.
Joseph is clumsy and accidentally dropped decoupling proteins into the plant he recently bought for his girlfriend. Unbeknownst to him, this might lead to a fight pretty soon... The uncoupling proteins were taken up by the plant's root cells, but did not localize anywhere else. Which of the following functions would be most disrupted by Joseph's mistake?

A. Lateral root formation

B. Cation exchange

C. Root hair formation

D. Polar transport of cytokinins

E. Perception of gravity

Ecology (1 Question)

Question 9.
A person arrives at a family reunion being held on his grandmother's yacht, located off the coast of Argentina. As he boards the yacht, it begins to tip dangerously when his extremely overweight uncle steps onboard. At this point, five family members are already on the yacht: your grandmother, your grandfather, your daughter, your niece, and your uncle. The grandmother and grandfather are no longer having children. All remaining individuals (daughter, niece, uncle) are assumed to have equal potential to survive and reproduce in the future. Without your help, the people on the yacht have a 30% chance of survival. If you do attempt to help them, their chances of survival increase but you have an 50% chance of dying in the process. Given these conditions, and assuming you want to maximize your genetic contribution to future generations, what is the minimum survival chance for the people on the yacht that would justify risking your life to help them?

A. 40%

B. 50%

C. 80%

D. 90%

E. You shouldn't help them at all

Ethology (1 Question)

Question 10.
Ian loves studying biology and loves penguins just as much. However, Ben the emperor penguin does not love Ian. In order to increase Ben's affection for him, Ian feeds Ben his favorite fish every time he comes close to him, which worked successfully. What type of conditioning is Ian performing on Ben?

A. Active Avoidance

B. Negative Punishment

C. Positive Punishment

D. Negative Reinforcement

E. Positive Reinforcement

Biosystematics (1 Question)

Question 11.
You're an avid explorer and biologist who went on an expedition last week to Antarctica. There you discovered a group of nonchalant baby emperor penguins that is shown below.



Which family of aves do the penguins belong to?

A. Corvidae

B. Anatidae

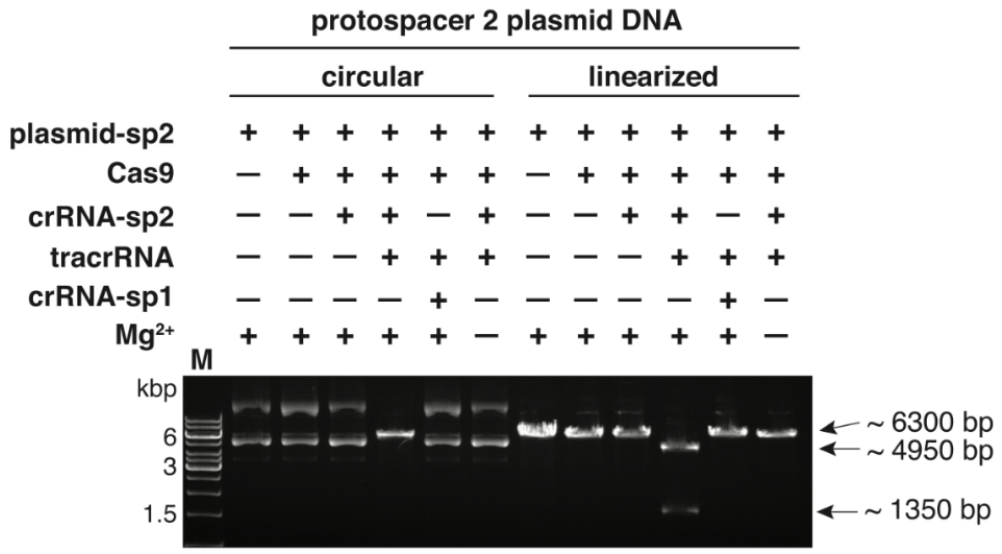
C. Spheniscidae

D. Trochilidae

E. Strigidae

Part B (MTF): 3 Questions

Question 12.
Shown below is data from a gel ran by Jennifer Doudna's lab during the study of the CRISPR-Cas9 system.



Based on the data above, which of the following statements is/are TRUE:

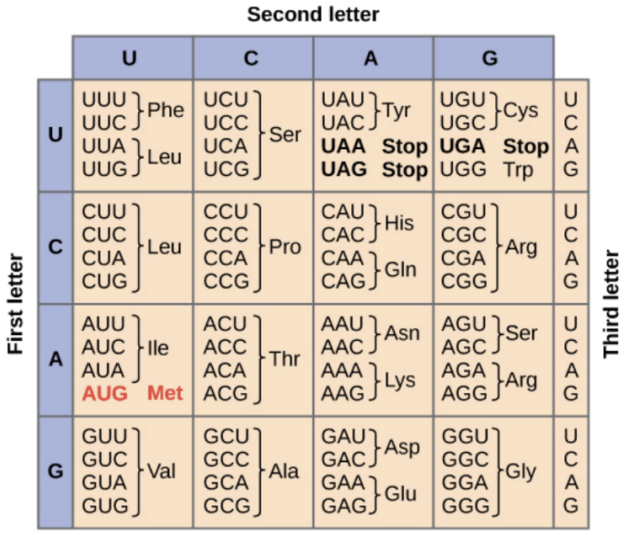
I. The upper bands seen in the circular DNA lanes correspond to nicked plasmids.

II. CRISPR-Cas9 can only cut circular DNA sequences.

III. Mg^{2+} is a required cofactor for Cas9 endonuclease activity.

IV. Both a spacer region and tracrRNA is required for Cas9 endonuclease activity.

Question 13.
Our DNA is organized into 3 letter codons, which code for different amino acids. Mutations in our DNA may result in impaired fidelity of DNA replication as well as improper protein translation. Which of the following statements regarding DNA and mutations is/are TRUE?



I. A 3 nucleotide deletion cannot result in a frame shift mutation.

II. If you were to introduce a single nucleotide deletion into a sequence, the likelihood of a nonsense mutation occurring along the open reading frame increases with sequence length.

III. A male honeybee would be more likely to utilize homologous recombination to repair a double stranded break, while a female honeybee would be more likely to utilize nonhomologous end joining.

IV. Cysteine's codon indicates that it is likely less commonly incorporated into proteins than serine is.

Question 14.
You isolate eight trp$^-$ mutants. You then perform a complementation test on these mutants. The data is shown below. In the table below, a (+) indicates growth on minimal media, a (-) indicates lack of growth on minimal media.

	m1	m2	m3	m4	m5	m6	m7	m8	WT
m1	-	+	+	+	+	+	+	+	+
m2		-	+	+	+	-	-	+	+
m3			-	+	-	+	+	+	+
m4				-	+	+	+	+	+
m5					-	+	+	+	+
m6						-	-	+	+
m7							-	+	+
m8								-	+

Indicate whether the following mutants are (true) or are not (false) mutants in the same genes.

I. m1, m2

II. m2, m6, m7

III. m3, m5

IV. m2, m8